Average / Instantaneous Rates of Change
(Section 2.1)

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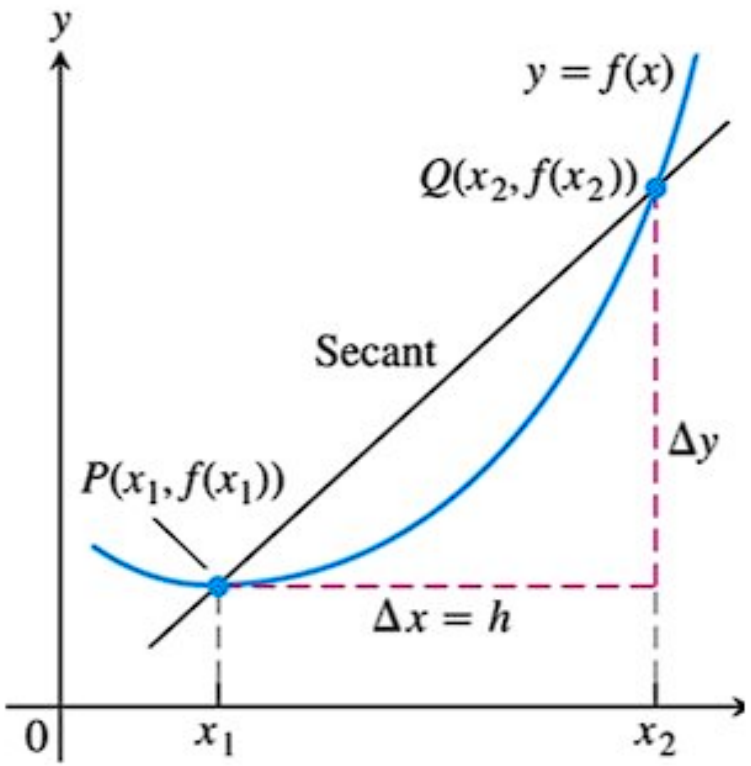
MATH 2413

Average Rate of Change

Definition

The **average rate of change** of a function over $t \in [x_1, x_2]$ is $\frac{f(x_2)-f(x_1)}{x_2-x_1}$.



Examples

Example

Usain Bolt ran 100 meters in 9.58 seconds. What was his average velocity during the sprint?

Examples

Example

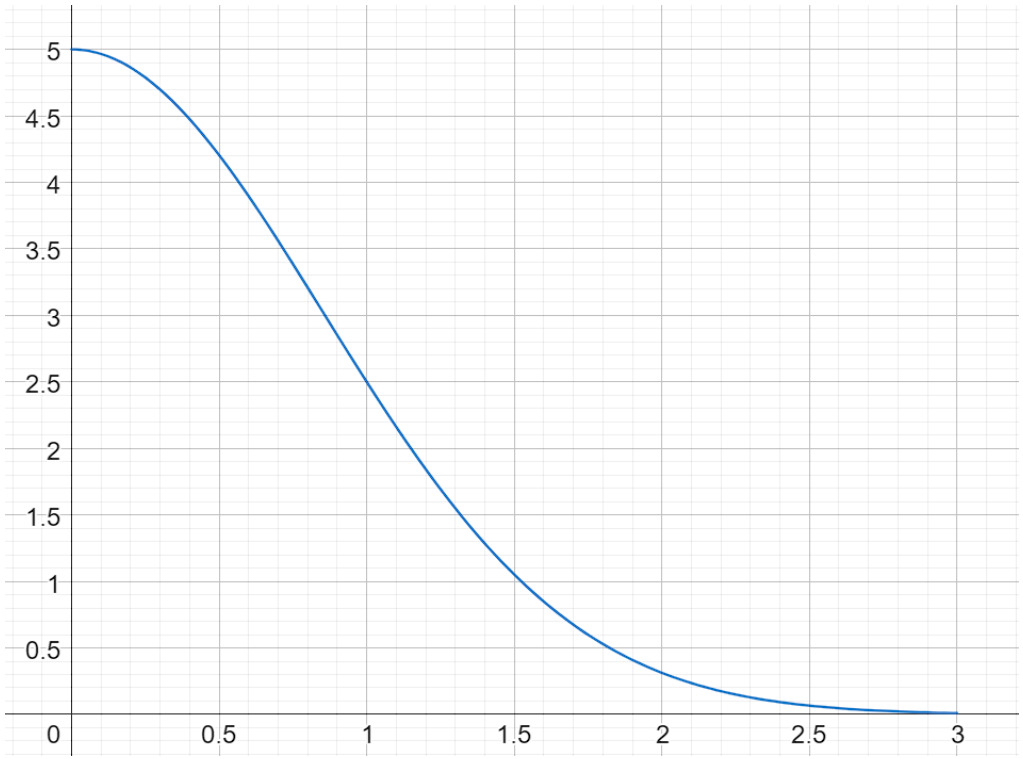
A car going from 0 to 60 mph in 4 seconds follows the distance function $f(t) = \frac{1}{480}t^2$, where t is measured in seconds and $f(t)$ is measured in miles. Find the average rate of change over the following intervals:

- $t = 1$ to $t = 2$
- $t = 0$ to $t = 4$
- $t = 3.99$ to $t = 4$

Examples

Example

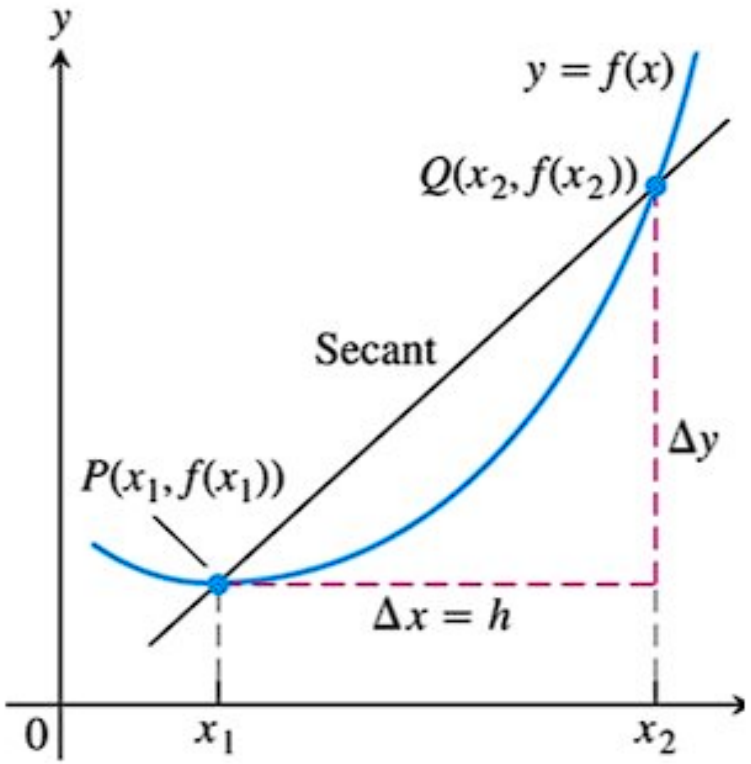
In the following graph, calculate the average rate of change between $t = 0.5$ and $t = 2.5$.



Average Rate of Change - Alternate Version

Definition

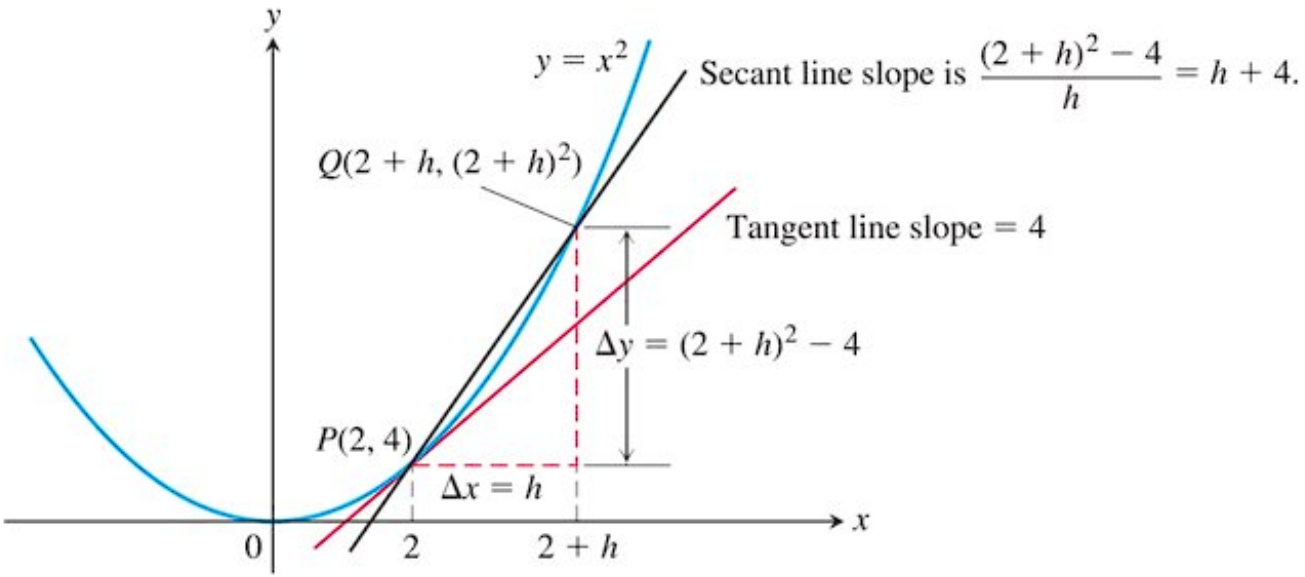
Suppose we let $h = x_2 - x_1$, or $x_2 = x_1 + h$. The ARoC formula changes to $\frac{f(x_1+h)-f(x_1)}{h}$.



Instantaneous Rate of Change

Definition

The **instantaneous rate of change** is the rate of change of a function at a single point, rather than between two points. This is calculated by moving x_2 infinitely close to x_1. In other words, we'll take the average rate of change formula and let $h \to 0$.



Examples

Example

Let $f(x) = x^2$. Find the ARoC between $x = 3$ and ...

- $x = 4$
- $x = 3.1$
- $x = 3.01$
- $x = 2$
- $x = 2.9$
- $x = 2.99$

Estimate the instantaneous rate of change of $f(x)$ at $x = 3$.

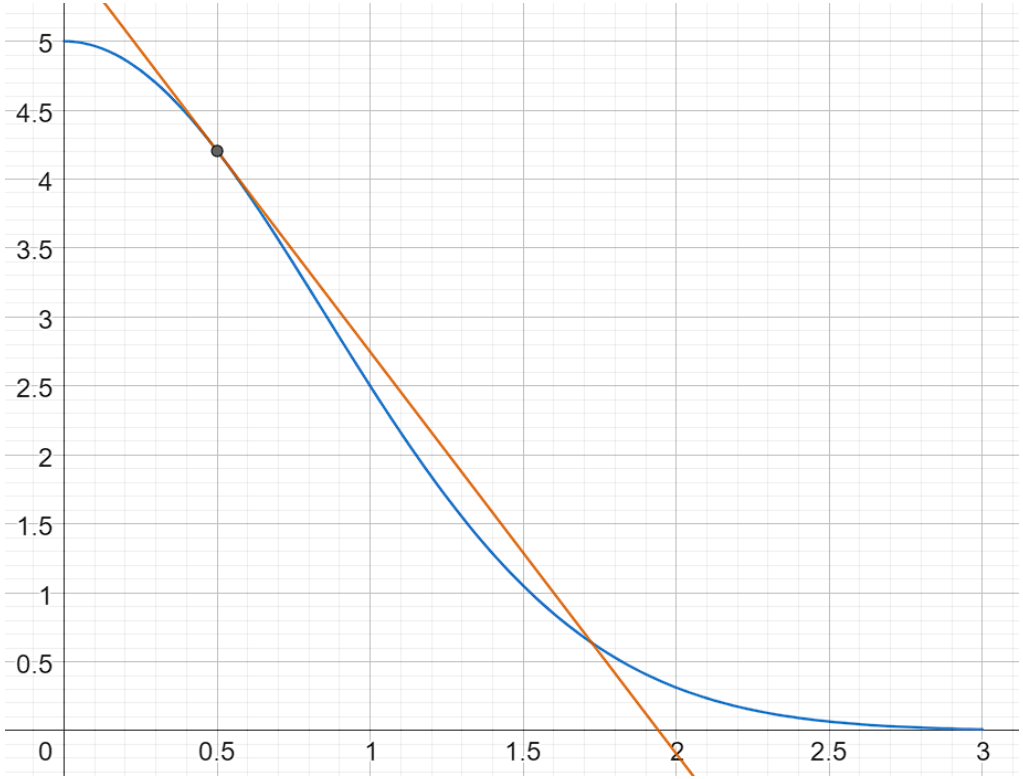
Examples

Example

For the following functions, find the average rate of change between $x = 1$ and $x = 1 + h$. Then find the instantaneous rate of change at $x = 1$.

- $f(x) = 3x^2$
- $g(x) = \sqrt{x}$
- $h(x) = \frac{1}{x}$

Examples

Example

For the graph below, estimate the instantaneous rate of change at $x = 0.5$.